11:20 AM

1️⃣ HOW THE SCENARIO QUESTION IS CHECKED

Examiners look for **MBSD concepts**, not real-world science accuracy.

They expect:

- Sensors → Microprocessor (8085/8086) → Interface → Output
- Use of **ADC, PPI, Memory, I/O devices**
- **Block diagram**
- **Step-wise working**

If you show this flow, **you get marks** even if the scenario changes.

2️⃣ UNIVERSAL MBSD BLOCK DIAGRAM (USE THIS EVERY TIME)

You can redraw this for **any scenario**:

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If you show this flow, **you get marks** even if the scenario changes.

2️⃣ UNIVERSAL MBSD BLOCK DIAGRAM (USE THIS EVERY TIME)

You can redraw this for **any scenario**:

Copy code

```
[ Sensors ]
     |
     ▼
[ Signal Conditioning ]
     |
     ▼
[ ADC ]
     |
     ▼
[ 8085 / 8086 Microprocessor ]
     |
     ├──► [ Memory (RAM/ROM) ]
     |
     ├──► [ 8255 PPI ]
     |          |
     |          ├──► Display (LCD/LED)
     |          └──► Alarm / Actuator
     |
     └──► [ Communication (Optional) ]
```

⚠️ **This diagram alone can fetch 40–50% marks.**

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⚠️ This diagram alone can fetch 40–50% marks.

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3️⃣ READY-TO-WRITE SCENARIO ANSWER

(Weather Forecasting System – MBSD Based)

Scenario Description

A weather forecasting system is designed to measure environmental parameters such as temperature, humidity, and pressure using sensors and process the data using a microprocessor-based system.

Block Diagram

(Draw the universal diagram above)

Working Explanation

1. **Sensors** measure weather parameters like temperature and humidity.
2. The analog signals from sensors are passed through **signal conditioning circuits** to remove noise.
3. An **ADC** converts analog signals into digital form.

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1. **Sensors** measure weather parameters like temperature and humidity.
2. The analog signals from sensors are passed through **signal conditioning circuits** to remove noise.
3. An **ADC** converts analog signals into digital form.
4. The digital data is sent to the **8085 microprocessor**.
5. The microprocessor processes the data using a stored program in **ROM** and temporary data in **RAM**.
6. **8255 PPI** interfaces output devices.
7. The processed data is displayed on an **LCD** or **LED display**.
8. If abnormal conditions occur, an **alarm** can be activated.

Conclusion

Thus, the weather forecasting system uses an MBSD architecture where sensors, ADC, microprocessor, memory and I/O devices work together to monitor and display weather conditions.

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4 AIR QUALITY MONITORING (SHORT VARIANT)

together to monitor and display weather conditions.

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4️⃣ AIR QUALITY MONITORING (SHORT VARIANT)

Just change **sensors + output meaning**.

Sensors: Gas sensor, dust sensor
Processing: Compare values with safe thresholds
Output: Display AQI level / buzzer alert

👉 **Same block diagram, same flow.**

5️⃣ HOW TO ADAPT TO ANY SCENARIO IN EXAM

When you see:

- **Weather** → temperature, humidity sensors
- **Air quality** → gas, smoke sensors
- **Agriculture** → soil moisture, temperature
- **Traffic** → IR sensors, counters

DO NOT panic.
Just write:

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processed by a microprocessor-based system using ADC, PPI, memory, and output devices."

When you see:

- **Weather** → temperature, humidity sensors
- **Air quality** → gas, smoke sensors
- **Agriculture** → soil moisture, temperature
- **Traffic** → IR sensors, counters

DO NOT panic.
Just write:

> "The system uses sensors whose output is processed by a microprocessor-based system using ADC, PPI, memory, and output devices."

🔑 MARKS-SCORING FORMULA (REMEMBER THIS)

Definition (2 lines)
➡ Block diagram
➡ Step-wise working (6–8 points)
➡ Conclusion

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1 Microprocessor vs Microcontroller

Microprocessor: A microprocessor is a CPU on a single chip that requires external components such as RAM, ROM, and I/O devices to form a complete system.

Microcontroller: A microcontroller is a complete computer on a single chip containing CPU, memory and I/O ports.

Differences:

	Microprocessor	Microcontroller
1	Needs external memory.	Memory is built-in.
2	Used in PCs.	Used in embedded systems

Example: 8086 is a microprocessor, while PIC is a microcontroller.

2) 8085 vs 8086

Feature	8085	8086	8088
Word size	8-bit	16-bit	16-bit
Address bus	16-bit	20-bit	20-bit
Max Memory	64 KB	1 MB	1 MB
Architecture	Simple	Pipelined	Pipelined
Speed	Slower	Faster	Moderate

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3- Handshaking:

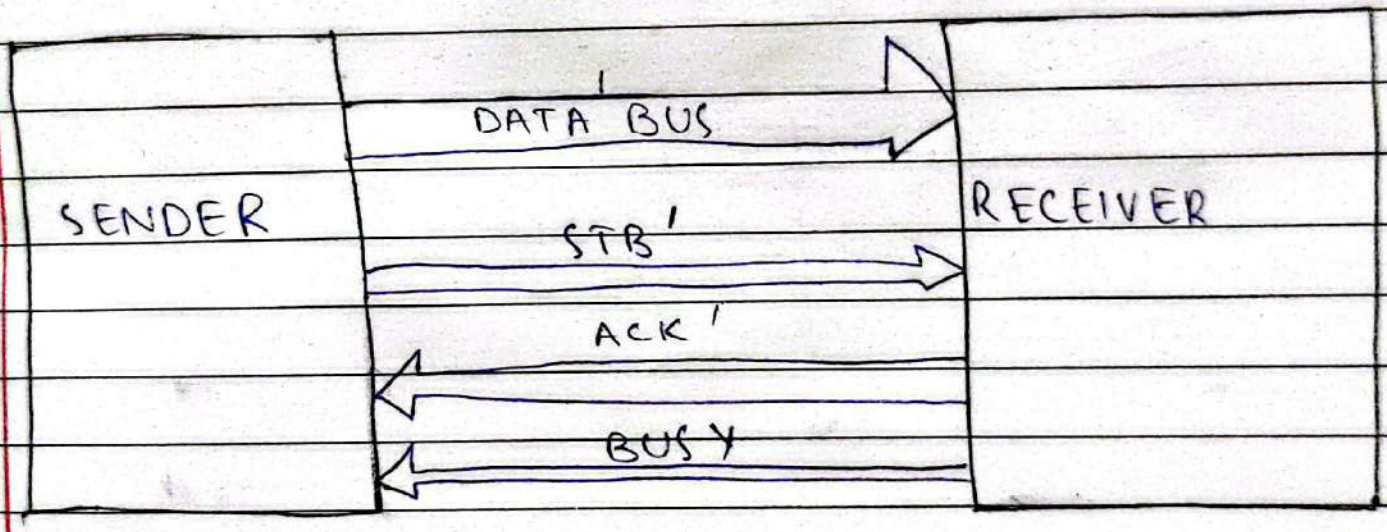
Handshaking is a method used to synchronize data transfer between CPU and peripheral devices.

Working:

1- CPU sends a request signal.
2- Peripheral checks readiness.
3- Peripheral sends acknowledgment.
4- Data transfer occurs.

Advantage:

Prevents data loss due to speed mismatch.



4 Programmable Peripheral Interface (8255)

8255 PPI is an interface device used to connect I/O devices with the microprocessor.

Ports: (A), (B), (C)

Modes:

0 → Simple I/O
1 → Handshaking I/O
(3) → Bi-directional

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7- Control Word Register:

Control Word Register configures the operation of 8255 ports.

Functions:

- Selects I/O Mode.
- Sets port direction.
- Controls Handshaking.

I/O Data Transfer Techniques:

I/O data transfer techniques are methods used to transfer data between the microprocessor and I/O devices based on speed requirements and CPU involvement.

There are three main I/O data transfer techniques

1) Programmed I/O

CPU Device —— Status Check ——→ I/O Device

CPU Device ←—— Data Transfer —— I/O Device

Explanation:

In programmed I/O, the CPU continuously checks the status of the I/O device. When the device becomes ready, data transfer takes place. The CPU remains busy during the entire process.

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2) Interrupt Driven I/O

I/O Device — Interrupt → CPU

CPU — ISR execution → Data Transfer.

Explanation:

The I/O device interrupts the CPU when it is ready. The CPU temporarily halts its current task, services the device, transfers data, and then resumes execution.

3) Direct Memory Access (DMA):

CPU — Initialization → DMA Controller

DMA Controller — Data Transfer → Memory

△

I/O Device

Explanation:

DMA allows direct data transfer b/w memory and I/O devices without continuous CPU involvement, making it the fastest technique.

Comparison

Technique	CPU Load	Speed
Programmed I/O	High	Slow
Interrupt I/O	Medium	Medium
DMA	Low	Fast

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Conclusion:

Among the three techniques, DMA is the fastest, while programmed I/O is the simplest. The choice depends on system requirements.

Program Data Transfer:

Unconditional Transfer: Data is transferred without any condition.

Conditional Transfer: Depends on flag status.

Double Buffering: Two Buffers used to avoid data loss.

Programmable Interrupt Controller:

8259 PIC manages multiple interrupt requests and reduces CPU burden.

Features:

Priority control

Interrupt Masking

Faster response.

Date:

STA Instruction:

STA (Store Accumulator) stores the content of accumulator into a specified memory location.

Operation:

1 - Fetch opcode
2 - Fetch address
3 - Store data

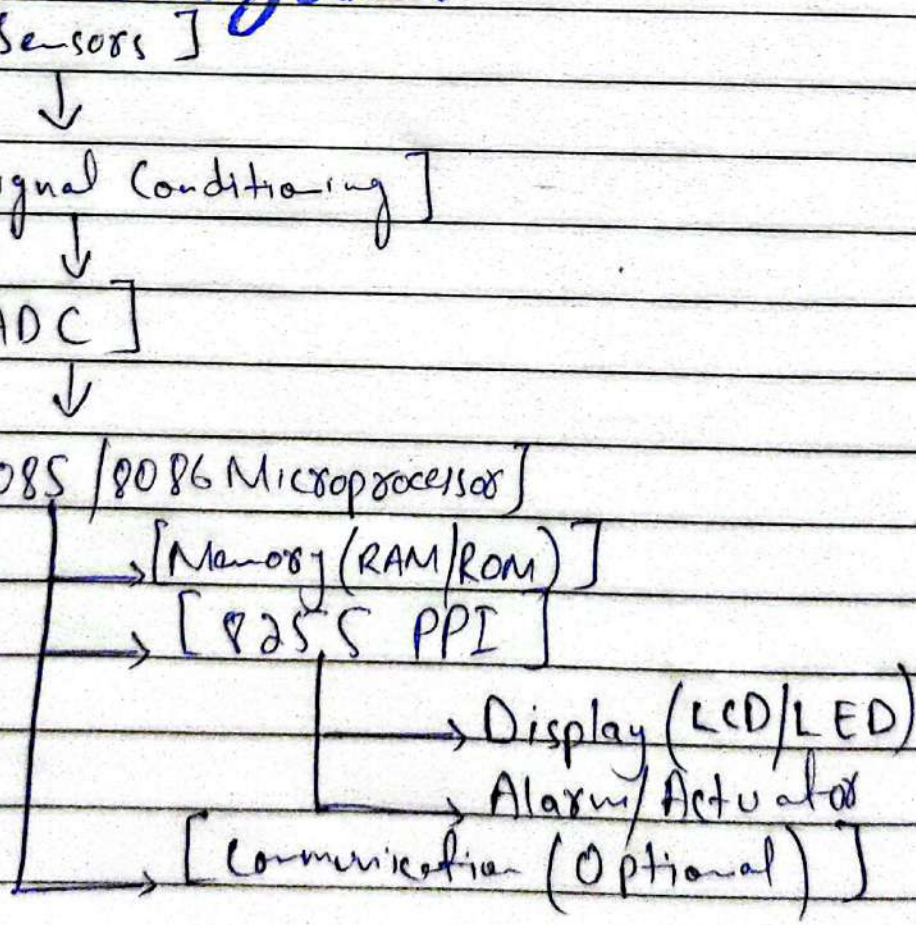
T - states:

Requires multiple machine cycles for execution.

(Timing Diagram if asked)

SCENARIO BASED Q:

Block Diagram:



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M050

Count the number of 1's in a number.

→ Count the no. of 1's of the content of register D and store the ~~content~~ count in register B.

```
MVI     B, 00H
MVI     C, 08H
MOV     A, D
BACK    RAR
JNC     SKIP
INR     B
SKIP    DCR, C
JNZ     BACK
HLT
```

2501 H = 04 H

2502 H = 34 H

2503 H = A9 H

2504 H = 78 H

2505 H = 56 H

Result = 2503 H = A9H

Largest of two numbers :-

~~The 16bit number a~~

```
LXIH    2501, H
MOV     A, M
INX     H
CMP     M          Compare
JNC     AHEAD      Number larger in accumulator go ahead
MOV     A, M
STA     2503
```

2501H = 98H

2502H = 87H

2503H = 98H

Smaller of two numbers:

```
LXIH    2501 H
MOV     A, M
INX     H
CMP     M
JC      AHEAD
MOV     A, M
STA     2503
HLT
```

2501 H = 84 H

2502 = 99 H

2503 H = 84 H

Calculate the sum of series of even number.

```
LDA     2500 H
MOV     C, A        (Initialize Counter)
MVI     B, 00H
LXI H   2501 H
BACK:   MOV A, M
ANI     01 # Mask bit 1 to 7 "
JNZ     SKIP
MOV     A, B
SKIP    INX H
DCR     C
JNZ     BACK
STA     2505 H
```